

for radiation from DC through the millimeter-wave region and into the infrared. The newly discovered high-temperature superconducting materials could revolutionise electronics as their very interesting properties and device behaviour would lead to an operating temperature where cooling requirements are greatly reduced and where hybrid semiconductor-superconductor circuits can be built that use the best features of each technology.

Electronic filters. A filter is an electronic circuit that selects some frequencies while blocking the rest. For instance, filters are used in radio sets to select the radio station users want to listen to; the radio set receives signals of all the stations, but the filter allows only the frequency of the selected station.

However, filters are not flawless; these always tend not to block a small part of the non-selected signals, especially if the latter use a frequency that is not very different from the selected one. This causes the very annoying situation of a radio picking up two stations at the same time, because the filter has trouble differentiating the two signals.

With the development of telecommunications over the last couple of years, the most efficient filters are required. Yet, the electric resistance of materials used to build these circuits limits their quality. Superconductors provide a solution to this problem.

Filters made of superconducting materials are more efficient than regular filters. These are used in cell phones' relay antennae, enabling them to pick up signals from sources at longer distances away. All over the world, several thousand relay antennae use superconducting filters. These filters could also be used in telecommunication satellites.

Superconducting circuits. These can be built so as to efficiently pro-

tect an electrical plant from power supply accidents (high voltage or short-circuit). These limiters react very quickly, guarantee an optimal protection of the plant and automatically reset when the accident is over.

Computer chips with superconducting circuits would be 50 to 100 times as energy-efficient as today's chips—an attractive trait given the increasing power consumption of the massive data centres that power the Internet's most popular sites. Superconducting chips also promise greater processing power.

Superconducting circuits that use so-called Josephson junctions have been clocked at 770 gigahertz, or 500 times the speed of the chip in iPhone 6. But Josephson-junction chips are big and complex to make; to top it all, these use such minute currents that their computation results are difficult to detect. So for the most part, they've been relegated to a few custom-engineered signal-detection applications.

Researchers have presented a new circuit design that could make simple superconducting devices much cheaper to manufacture. And while these circuits' speed probably wouldn't topple today's chips', they could solve the problem of reading out the results of calculations performed with Josephson junctions.

SQUID applications. In superconducting electronics, Josephson junctions and SQUIDs are active devices. Analogue electronics is used for detectors, mixers, voltage standards and ultra-sensitive magnetometers with their application to fields ranging from radio astronomy through metrology to stand-off security imaging.

RSFQ electronics. Rapid single-flux quantum (RSFQ) electronics is currently under development. It is a binary sort of electronics that uses 0s and 1s (contrary to quantum electronics). Instead of silicon transistors, RSFQ electronics uses

Josephson junctions to manipulate magnetic flux quantum. Flux quantum is used to encode bits (0 or 1). This technique allows operation at a very fast speed, up to 100GHz, while dissipating much less heat than average circuits.

RSFQ electronics represents a potential leap for electronics, but, besides low temperatures, it also requires a radical change of technology, which significantly delays any marketing possibility. Superconducting digital logic circuits use single-flux quanta (SFQ), also known as magnetic flux quanta, to encode, process and transport data. While voltages and capacitors are important in semiconducting logic circuits such as CMOS, currents and inductors are the most important parameters in SFQ logic circuits.

Research in superconducting electronics is mostly centered on RSFQ digital electronics. Associated digital circuits use Josephson tunnel junction as the basic component. RSFQ circuits use Josephson junctions that are resistively shunted in a careful way in order to be able to generate, propagate and store binary information through quanta of magnetic flux. This binary information propagates through the circuit as quantised voltage pulses. The pulse duration being of the order of a few picoseconds, the repetition rate of such pulses—the maximum clock frequency that RSFQ circuits can handle—is several tens to several hundreds of GHz. Pulse propagation takes place without losses in the superconducting circuits at a speed close to half the light speed.

The presence of a quantised pulse in a given time interval is digitally coded as '1', while the absence of such pulse is coded as '0'. Binary information is stored in the circuits as a permanent current running in a superconducting loop. The flux associated with this current corresponds to the magnetic flux quantum. **EFY**